Not All Models Are Inflated: Profiling Contamination Dose-Response Across Model Families

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Abstract

Benchmark contamination is widely assumed to inflate language model scores, yet the actual effect has never been systematically measured across model families. We perform controlled SFT injection at five dosage levels into six models spanning five independent families (7B–14.7B parameters) and discover that contamination produces not one effect but a spectrum of dose-response patterns. Benefit (+2.4pp mean) and Immune ($\Delta <$ 1 percentage point [pp]) are replicated across independent families (three and two, respectively); more extreme responses—catastrophic collapse (-9 to -25pp across seeds) and V-recovery—are each observed in one family (Qwen2.5) across three seeds, requiring further replication. In our six-family sample, contamination vulnerability concentrates in the 65–70% MMLU baseline band; models at extreme baselines show limited sensitivity. Counter-intuitively, the 14B Qwen model is 3$\times$ more fragile than 7B. These effects are benchmark-conditional: MMLU-targeted contamination improves GSM8K in five of six families ($+1$ to $+16$pp), even in models harmed on MMLU, providing converging evidence for capability elicitation rather than item memorization. Difference-in-differences analysis finds no item-specific memorization on MMLU (minimum detectable effect size 1.2–2.0pp); ARC has limited power (2.3–5.6pp). Profiles are robust: within-seed bootstrap stability exceeds 99.6% for 4/5 MMLU families, cross-seed direction agreement reaches 5/6, though strict sign consistency is 3/6. ARC-Challenge achieves 4/5 directional agreement despite its smaller item pool (1,172 vs. 14,042 items). We characterize contamination detection signal failure: Min-K%++ exhibits directional reversal at small scales (paired Cohen's $d = -1.98$ at d100; $|d| > 1.4$ at d005), actively misleading practitioners. Simulation establishes that IRT-based contamination diagnosis requires ensemble sizes ($J \geq 50$) above current practice. Out-of-sample validation on a held-out sixth family (LLaMA-3.1-8B) confirms the profiles generalize: it is classified Immune despite a baseline matching OLMo-2's Benefit, showing that model family characteristics beyond baseline ability modulate vulnerability. Our findings demonstrate that "contamination equals inflation" masks substantial heterogeneity; model-specific dose-response profiling is necessary for reliable benchmark interpretation.

1 Introduction

Benchmark scores drive model selection and research priorities for large language models, yet training data contamination is pervasive: trillion-token web crawls routinely ingest benchmark items (Oren et al., 2024), and practitioners cannot determine how much of a reported score reflects genuine capability versus memorization (Hendrycks et al., 2021; Chang et al., 2024). Despite twenty mitigation strategies (Sun et al., 2025), contamination-free benchmarks (White et al., 2024), and dynamic regeneration (Dong et al., 2024), a fundamental question remains: what is the actual effect of contamination on aggregate scores? The implicit assumption is uniform inflation, but this has never been tested across model families.

We perform controlled SFT injection (LoRA, five dosage levels) into six models spanning five independent families from five labs (Alibaba, Microsoft, Allen AI, Google, TII) and discover a spectrum of dose-response patterns (Figure 1)—from immunity (Phi-4, $\pm$0.5pp) and benefit ($+3$–4pp, three families) to collapse in the 65–70% baseline band (Qwen-14B, -9 to -25pp across seeds).

In our six-family sample, contamination vulnerability concentrates in the 65–70% MMLU baseline band: below it, models benefit (OLMo-2 $+3.2$pp, Falcon3 $+4.2$pp, Gemma-2 $+3.8$pp);

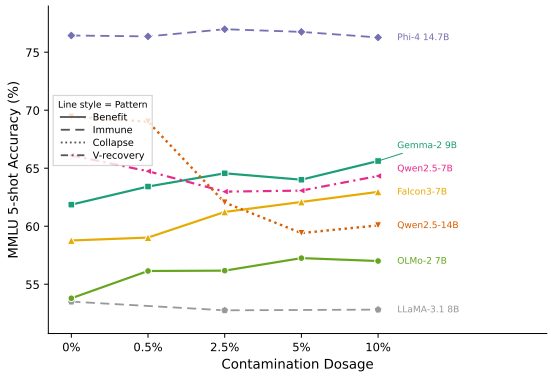


Figure 1: MMLU accuracy vs. contamination dosage (5-shot, seed=42; LLaMA-3.1 at seed=0 with d000/d025/d100). Vulnerability concentrates in the 65–70% baseline band. Qwen-14B is 3× more fragile than 7B. Cross-seed patterns in Table 11; ARC in Appendix B.

above it, they are harmed or immune. Counterintuitively, the 14B model is 3× more fragile than 7B, and these effects are benchmark-conditional: the same model can collapse on MMLU (-9.4pp) yet gain on GSM8K ($+13$pp). ContamCal, our two-stage pipeline combining multi-signal item filtering with contamination-augmented IRT (2PL+γ_k), enables this characterization.

Our contributions:

1. **Dose-response profiles with cross-seed and cross-benchmark validation.** Controlled injection across five families reveals a spectrum of dose-response patterns; in our six-family sample, vulnerability concentrates in the 65–70% MMLU baseline band. Out-of-sample validation on LLaMA-3.1-8B confirms generalizability. Cross-benchmark transfer reveals three family-specific patterns on GSM8K (null, consistent, directional reversal). Multi-seed evaluation ($n = 3$) shows direction is reproducible but magnitude is seed-sensitive. Domain analysis reveals within-benchmark heterogeneity (Social Sciences $P(\gamma_d < 0) = 99.7\%$).

2. **No detectable item-specific memorization at current power levels.** Difference-in-differences analysis on 14,042 MMLU items finds no differential accuracy gain for contaminated vs. clean items in any model family (MDES 1.2–2.0pp on MMLU; ARC power is limited at 2.3–5.6pp). Results are consistent with global capability modulation, challenging the assumption that contamination operates primarily through item-specific memorization.

3. **Signal failure characterization.** Min-K%++ (Zhang et al., 2025) exhibits directional reversal at 0.5B (paired Cohen's $d = -1.98$ at d100; $|d| > 1.4$ across benchmarks at d005). RRF-guided item filtering is a prerequisite for IRT convergence ($\Delta\tau = 0.18$ vs. size-matched random). Simulation establishes γ recovery requires $J \geq 50$ at AUC ≥ 0.55, above current practice ($J = 8$–30).

2 Related Work

Contamination detection and mitigation. Detection methods flag contaminated items via token-level surprisal (Shi et al., 2024; Zhang et al., 2025), accuracy gaps on paraphrases or rephrased items (Dekoninck et al., 2024; Yang et al., 2023), kernel divergence (Choi et al., 2025), completion probes (Dong et al., 2024; Golchin and Surdeanu, 2024), and parallel test sets (Zhang et al., 2024); see Chen et al. (2025); Singh et al. (2024); Sainz et al. (2023) for surveys. Balloccu et al. (2024) document contamination malpractices in closed-source LLMs; Bordt et al. (2025a) argue that contamination effects are often negligible in practice. Earlier work characterized contamination as memorization (Carlini et al., 2023; Magar and Schwartz, 2022; Li and Flanigan, 2024; Xu et al., 2024); Tao et al. (2025) and Kocyigit and Yildirim (2025) address RL post-training. These methods perform binary classification without quantifying effect sizes. On mitigation, Sun et al. (2025) find none of twenty approaches reliably restores clean performance; DCR (Xu et al., 2025) estimates decontaminated scores but without uncertainty quantification. ContamCal consumes detection signals as inputs and produces calibrated ability with credible intervals. Three concurrent lines of work directly address contamination quantification. Han et al. (2026) apply observational DIF to detect contamination without controlled dosage, achieving detection but not dose-response profiling. Schaeffer et al. (2026) perform controlled pretraining-stage injection, establishing causal contamination effects at the pretraining level but with different mechanics than SFT-stage leakage. Our work complements both: like Schaeffer et al., we use controlled injection for causal identification, but at the SFT stage with precise dosage control; like Han et al., we employ

IRT-based modeling, but with ground-truth contamination labels enabling dose-response characterization.

IRT for LLM evaluation. Item Response Theory (Lord, 1980; Hambleton et al., 1991) has gained traction for LLM evaluation (Liang et al., 2023): Rodriguez et al. (2021), TinyBenchmarks (Polo et al., 2024), Growing Pains (Habba et al., 2026), HiBayES (Luettgau et al., 2025), PSN-IRT (Zhou et al., 2026); see Lalor et al. (2024) for a tutorial and Gu et al. (2024) for evaluation standardization. Most directly related, Han et al. (2026) apply group-based DIF (Holland and Wainer, 1993) to quantify item-level contamination effects. ContamCal differs by using continuous exposure covariates (explanatory DIF; De Boeck and Wilson, 2004), providing calibrated ability with credible intervals, and domain-level diagnosis (γ_d); the approaches are complementary (item-level DIF for benchmark curation, model-level calibration for score interpretation). Our γ parameter is formally an instance of explanatory DIF (De Boeck and Wilson, 2004; Eckerly, 2017): the contamination covariate differentially affects response probability, analogous to item compromise models in educational measurement (Sinharay, 2017; Holland and Wainer, 1993; Camilli and Shepard, 1994; Swaminathan and Rogers, 1990). We use "calibration" in the psychometric sense of adjusting scores to approximate uncontaminated performance, distinct from probability calibration (Guo et al., 2017).

3 Method

ContamCal operates in two stages: Stage 1 estimates per-item contamination exposure from multiple detection signals via training-free fusion; Stage 2 embeds these scores in a contamination-augmented IRT model to produce calibrated ability estimates with credible intervals.

3.1 Problem Formulation

Let $y_{ijk} \in \{0, 1\}$ denote the binary response of model j ($j = 1, \ldots, J$) to item i ($i = 1, \ldots, I_k$) on benchmark k. Standard IRT assumes $P(y_{ijk} = 1) = f(\theta_j, \boldsymbol{\xi}_i)$ where θ_j is latent ability and $\boldsymbol{\xi}_i$ collects item parameters. When training corpora contain benchmark items, the observed ability $\hat{\theta}_j$ confounds genuine capability with memorization.

Our goal is to estimate θ_j^{clean}, the ability each model would exhibit absent contamination, and to diagnose contamination effects at domain granularity. We adapt item compromise IRT (Eckerly, 2017; Sinharay, 2017) and explanatory IRT (De Boeck and Wilson, 2004) for LLM evaluation: replacing binary leakage indicators with continuous exposure scores $e_{ij} \in [0, 1]$, fitting per-benchmark models to handle the shared item pool (San Martín et al., 2009), and recovering clean ability via counterfactual inference (setting $e_{ij} = 0$).

3.2 Stage 1: Multi-Signal Exposure Mapping

For each model j and item i, we consider four candidate detection signals. **S1** (Embedding similarity): cosine similarity between the item embedding and its nearest neighbor in the SFT training corpus. **S2** (Min-K%++): a membership inference statistic based on token-level surprisal (Zhang et al., 2025). **S3** (ConStat): the accuracy gap between original and paraphrased item variants (Dekoninck et al., 2024). **S4** (Self-Critique): negative entropy of model self-evaluation distributions, inspired by the finding that LLMs can assess their own knowledge (Kadavath et al., 2022); concurrent work by Tao et al. (2025) uses a related but mechanistically distinct entropy collapse approach targeting RL-trained models. S3 requires paired paraphrased items as reference and is therefore limited to benchmarks with existing paraphrase sets. Our primary pipeline uses the remaining three signals ($\mathcal{S} = \{\text{S1}, \text{S2}, \text{S4}\}$) and applies to any benchmark without auxiliary data. S2 and S4 exhibit known failure modes under RL post-training, which we characterize in §5.6. Item-level classification performance of the exposure scores is reported in Appendix AA.

We aggregate signals via Reciprocal Rank Fusion (RRF; Cormack et al., 2009):

$$e_{ij} = \sum_{s \in \mathcal{S}} \frac{1}{\kappa + r_s(i)}, \tag{1}$$

where $r_s(i)$ is the rank of item i under signal s (rank 1 = most suspicious), $\kappa = 60$ is the standard RRF smoothing constant, robust to near-random component signals, and $|\mathcal{S}| = 3$ in our primary configuration. Raw scores are min-max normalized per model to $e_{ij} \in [0, 1]$; the distribution is highly right-skewed (mean 0.041).

3.3 Stage 2: Contamination-Augmented IRT

Primary model: 2PL+γ. Our primary specification augments the two-parameter logistic (2PL)

model (Lord, 1980; Hambleton et al., 1991) with a contamination covariate. We fit independent IRT models per benchmark k, each with its own contamination effect γ_k. Per-benchmark fitting avoids cross-benchmark item parameter borrowing while permitting heterogeneous contamination effects. For benchmark k, the response probability is:

$$P(y_{ijk} = 1) = \sigma\big(a_i(\theta_j - b_i) + \gamma_k \cdot e_{ij}\big), \quad (2)$$

where $\sigma(\cdot)$ is the logistic function, $a_i > 0$ is item discrimination, $b_i \in \mathbb{R}$ is item difficulty, θ_j is model ability, and γ_k is the scalar contamination effect for benchmark k. Each benchmark is fitted independently; when finer-grained diagnosis is needed, we replace γ_k with per-domain effects γ_d ($d = 1, \ldots, D$), fitting Eq. (2) separately for each cognitive domain within a benchmark (§5.7).

We place γ_k outside the item discrimination bracket, modeling contamination as a uniform logit shift (De Ayala, 2009) rather than the non-uniform DIF form that would scale contamination effects with item discrimination. For MC-format benchmarks, a 4PL extension adds guessing and ceiling asymptotes (Appendix G); this requires $J \geq 16$ and is not used in domain-level analyses ($J = 13$).

Inference. We perform Bayesian inference via NUTS (Hoffman and Gelman, 2014) in Stan (Carpenter et al., 2017) with four chains and target acceptance 0.95. Priors: $\theta_j \sim \mathcal{N}(0, 1)$, $a_i \sim \text{LogNormal}(0, 0.5)$, $b_i \sim \mathcal{N}(0, 1)$, $\gamma_k \sim \mathcal{N}(0, 1)$. Convergence requires split-$\hat{R} < 1.01$ and bulk ESS > 400 for all parameters. Model fit is assessed via the posterior predictive p-value (PPP), where values near 0.5 indicate adequate fit.

Calibrated score. The decontaminated accuracy for model j on benchmark k is the posterior expectation under zero exposure:

$$\text{Acc}_j^{\text{cal}} = \frac{1}{I_k} \sum_{i=1}^{I_k} \mathbb{E}_{\boldsymbol{\psi} \sim p(\boldsymbol{\psi}|\mathbf{y})} \left[\sigma\big(a_i(\theta_j - b_i)\big)\right], \quad (3)$$

with credible intervals from posterior quantiles. In practice, we average $\sigma(a_i(\theta_j - b_i))$ over posterior draws with $e_{ij} = 0$; the spread of draws yields the credible interval. Models with noisier exposure signals receive wider intervals by construction.

Identifiability. The reliability of γ depends on signal AUC and ensemble size J; simulation validates that recovery requires AUC$\geq$0.55 with $J \geq 50$ (Appendix F). Empirically, exposure scores e_{ij} and item difficulty b_i are near-orthogonal (Pearson $|r| < 0.13$, $R^2 < 0.014$ across benchmarks), confirming that γ is identified from dosage-induced variation rather than confounded with difficulty. Eq. (3) depends on θ_j, not γ directly, predicting that vanilla 2PL ($\gamma = 0$) achieves comparable calibration to 2PL+γ, with γ_k serving diagnosis rather than calibration (§5.5).

3.4 Dose-Response Profile Classification

Independent of the IRT calibration pipeline, we formalize how each model's accuracy trajectory across dosage levels is classified into a canonical dose-response pattern, drawing on the dose-response modeling tradition in pharmacology (Hill, 1910; Ritz et al., 2015). Algorithm 1 defines a deterministic decision tree over the accuracy vector $\mathbf{a} = (a_0, a_5, a_{25}, a_{50}, a_{100})$, parameterized by noise threshold τ, severity threshold σ, saturation ratio ρ, and late-gain floor λ. The top-level split is governed by $\Delta = a_{100} - a_0$ relative to τ; within-branch refinements distinguish Collapse from V-recovery via mid-dosage trough detection. A mid-trajectory volatility check (v^*) distinguishes true immunity (Phi-4: $v^* = 0.34$pp) from V-recovery with full endpoint recovery (Qwen-7B seed=123: $v^* = 3.22$pp $> \tau$). The default $\tau = 2$ pp corresponds to approximately one standard deviation of cross-seed accuracy variability. All thresholds are explicit and adjustable; a 30-combination grid sweep confirms all families are stable across all (τ, σ) combinations (180/180 model–seed–threshold pairs; Appendix S).

Family-level aggregation. Algorithm 1 classifies individual seed trajectories. Family-level classification uses *direction* as the primary criterion (§5.3): Immune if the bootstrap CI of the cross-seed mean Δ(d100) includes zero and |mean Δ| $< \tau$; Borderline if the CI includes zero but |mean Δ| $\geq \tau$; otherwise Harm/Benefit by sign, with per-seed shape providing supplementary characterization. Parametric model selection (Appendix Z) confirms Algorithm 1's classifications for extreme effects but lacks power to resolve moderate patterns at $n = 3$ seeds, motivating the per-seed rule-based approach. Our methodological contribution lies not in the IRT specification—which follows standard explanatory IRT practice—but in the controlled SFT injection paradigm that enables causal dose-response

Algorithm 1 Dose-Response Profile Classification

Require: Accuracy vector $\mathbf{a} = (a_0, a_5, a_{25}, a_{50}, a_{100})$ at dosages $d \in \{0, 5, 25, 50, 100\}\%$
Require: Noise threshold $\tau = 2$ pp, severity threshold $\sigma = 5$ pp
Require: Saturation ratio $\rho = 0.3$, late-gain floor $\lambda = 0.5$ pp
Ensure: Pattern $p \in \{\text{IMMUNE}, \text{COLLAPSE}, \text{V-RECOVERY}, \text{BENEFIT}\}$

$\Delta \leftarrow a_{100} - a_0$ ▷ end-to-end accuracy shift
$m^* \leftarrow \min(a_5,\ a_{25},\ a_{50})$ ▷ mid-dosage minimum
$v^* \leftarrow \max_{i \in \{5,25,50\}} |a_i - a_0|$ ▷ max mid-trajectory deviation
if $|\Delta| \le \tau$ **then**
  if $v^* \le \tau$ **then**
    return IMMUNE ▷ fluctuation within noise band
  else if $m^* < a_0$ **and** $m^* < a_{100}$ **then**
    return V-RECOVERY ▷ V-dip with full endpoint recovery
  else
    return IMMUNE ▷ non-V volatility, endpoint stable
  end if
else if $\Delta < -\tau$ **then**
  if $\Delta \le -\sigma$ **then**
    return COLLAPSE ▷ severe monotonic degradation
  else if $m^* < a_0$ **and** $m^* < a_{100}$ **then**
    return V-RECOVERY ▷ mid-dosage trough with partial recovery
  else
    return COLLAPSE ▷ moderate decline, no V-shape
  end if
else ▷ $\Delta > \tau$
  $g_1 \leftarrow a_{25} - a_0$ ▷ first-half gain
  $g_2 \leftarrow a_{100} - a_{25}$ ▷ second-half gain
  $\ell \leftarrow a_{100} - a_{50}$ ▷ late gain
  if $(g_1 > 0$ **and** $g_2 < \rho \cdot g_1)$ **or** $\ell < \lambda$ **then**
    return BENEFIT ▷ saturating gain
  else
    return BENEFIT ▷ sustained gain
  end if
end if

profiling of contamination effects.

4 Experimental Setup

4.1 Controlled SFT Injection

We construct a controlled contamination testbed spanning five independent model families from five labs, trained on independent corpora with distinct pre-training recipes: Qwen2.5 (Yang et al., 2024) (0.5B, 1.5B, 3B, 7B, 14B), Phi-4 (Abdin et al., 2024) (14.7B), OLMo-2 (Team OLMo et al., 2025) (7B), Gemma-2 (Gemma Team, 2024) (9B), and Falcon3 (Technology Innovation Institute, 2024) (7B). All models are fine-tuned with a unified LoRA configuration (Hu et al., 2022) ($r = 16$, $\alpha = 32$, dropout$= 0$, targeting all attention and MLP projections; lr$= 5 \times 10^{-5}$, bf16, 1 epoch, batch size$= 1$, gradient accumulation$= 16$, warmup ratio$= 0.03$; trainable parameters range from 0.45% of Phi-4 14.7B to 2.10% of Qwen2.5-0.5B) on a general-purpose SFT corpus into which benchmark items are injected at five dosage levels: d000 (0%, clean control), d005 (5%), d025 (25%), d050 (50%), and d100 (100%). Contaminated items augment (not replace) the base SFT corpus, yielding 50.0k (d000), 50.8k (d005), 54.1k (d025), 58.3k (d050), and 66.5k (d100) total training examples (+33% volume at d100 vs. d000).

Qwen2.5 serves as the primary within-family series (five scale points, 0.5B–14B). Models at $\le$3B are trained with two seeds at four dosage levels; 7B and 14B at all five dosages. The three cross-family models are each trained at five dosages with a single seed, serving as consistency checks rather than independent confirmatory analyses. This yields over 55 checkpoints spanning eight sizes across five families. We exclude 0.5B from MC IRT due to position bias (A-option rate: 46% at d000 to 98% at d100; §5.6). We additionally validate our dose-response profiles on a held-out sixth family, LLaMA-3.1-8B (Grattafiori et al., 2024) (Meta), which was not used during profile construction. LLaMA-3.1 is trained at three dosage levels (d000, d025, d100) to verify that Algorithm 1 generalizes to unseen model families.

4.2 Benchmarks, Domains, and Evaluation

We evaluate on three benchmarks spanning distinct cognitive demands and response formats.

MMLU (Hendrycks et al., 2021) (5-shot): 14,042 multiple-choice items across 57 subjects, grouped into 8 cognitive domains (Math, Science, CS & Engineering, Medical, Humanities, Social Sciences, Business & Law, Other). This stratifica-

tion reveals within-benchmark heterogeneity that aggregate scoring obscures (§5).

ARC-Challenge (Clark et al., 2018) (25-shot): 1,172 multiple-choice science reasoning items.

GSM8K (Cobbe et al., 2021): 1,319 open-ended math problems whose free-form generation format avoids multiple-choice artifacts. Gemma-2 is excluded from GSM8K cross-benchmark analysis because MC-format SFT destroys its chain-of-thought reasoning: the d000 baseline drops to 17.8% (vs. 73–84% for other families), indicating CoT collapse rather than a contamination effect.

All multiple-choice evaluations use logprob-based scoring under an aligned instruction-template format with fixed few-shot counts; all 75 model–dosage configurations are evaluated under this identical protocol.

We assess calibration quality via MAE (calibrated vs. ground-truth clean accuracy), Kendall τ (rank preservation), credible interval coverage, posterior predictive p-value (PPP), γ direction ($P(\gamma < 0 \mid \text{data})$), and split-$\hat{R}$ convergence.

4.3 Baselines

We compare against seven baselines: **B0-Raw** (no adjustment), **B1-Remove&Rescore** (drop flagged items; B1b uses RRF-filtered items), **B2-Single2PL+γ** (aggregate γ without domain stratification), **B3-Vanilla2PL** ($\gamma = 0$, isolating IRT ability estimation), **B4-LinearRegression** (residuals on exposure scores), **B5-Emperor** (mitigation strategies from Sun et al. (2025); B5b uses the best-performing variant), and **B6-DCR** (our reimplementation following Xu et al., 2025).

5 Results

Cross-family results (§5.1) use seed=42 data across six models from five independent families. Within-family IRT analyses (§5.5–5.7) use $J = 16$ Qwen2.5 models (8×1.5B + 8×3B); domain-level contamination estimates γ_d (§5.7) are reported at $J = 16$ in the text and $J = 13$ in Figure 11.

5.1 Dose-Response Patterns

Applying Algorithm 1 to seed=42 accuracy trajectories yields a spectrum of dose-response patterns (Table 1, Figure 1), not a uniform inflation effect.

Baseline ability gradient. In our six-family sample, contamination vulnerability concentrates in the 65–70% MMLU baseline band (Table 1). Below 62%, three Benefit families gain +3.2 to +4.2pp; in the vulnerable band, Qwen-7B shows V-shaped harm-then-recovery while Qwen-14B collapses (−9.4 to −25.0pp across seeds)—3× more fragile despite higher baseline—this within-family divergence may reflect catastrophic interference at larger scale. Above 76%, Phi-4 is immune. ARC direction is consistent for 5/6 models (Appendix B); domain-level effects vary up to 1.8× (Appendix T).

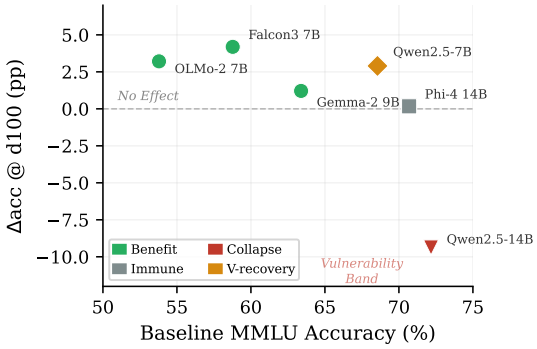


Figure 2: Baseline MMLU accuracy vs. contamination response at d100. The 65–70% band exhibits the most heterogeneous responses.

Table 1: Observed dose-response patterns under controlled SFT injection (seed=42, MMLU). Δacc: d000→d100 change (95% CI from item-level paired bootstrap, $B = 10{,}000$). Stab.: bootstrap classification stability. Classifications are robust to threshold hyperparameter choice (5/5 original families at 100% stability across 30 reasonable (τ, σ) combinations; Appendix S).

Pattern	Model	d000 Acc.	Δacc	95% CI	Peak	Stab.
Immune	Phi-4 14.7B	76.4%	±0.5pp	[−0.60, +0.26]	negl.	100%
Collapse	Qwen-14B	69.4%	−9.4pp‡	[−12.74, −9.0]	—	100%
V-recovery	Qwen-7B	66.1%	−1.8pp	[−2.36, −1.22]	-3.1^{d025}	99.6%
Benefit	Gemma-2 9B	61.9%	+3.8pp	[+3.18, +4.33]	mono.	76.6%
Benefit	Falcon3 7B	58.8%	+4.2pp	[+3.60, +4.79]	mono.	96.1%
Benefit	OLMo-2 7B	53.8%	+3.2pp†	[+2.73, +3.71]	sat.^{d050}	100%

‡Cross-seed range −9.4 to −25.0pp (s42/s123/s0); see Table 11.
†Seed-sensitive: s42 +3.2pp, s123 −0.6pp, s0 +1.4pp; cross-seed CI crosses zero. See §5.3.

Out-of-sample validation. Out-of-sample validation on LLaMA-3.1-8B (Meta; d000, d025, d100) yields Immune on both MMLU (−0.69pp) and ARC (+0.09pp), despite matching OLMo-2's Benefit baseline (53.5% vs. 53.8%)—demonstrating family-level factors beyond baseline ability modulate vulnerability. Two of four patterns replicate across families: Immune (Phi-4, LLaMA-3.1), Benefit (OLMo-2, Falcon3, Gemma-2)—the most frequently observed pattern. LLaMA-3.1 shows the strongest GSM8K transfer (+16.4pp; Appendix C), demonstrating

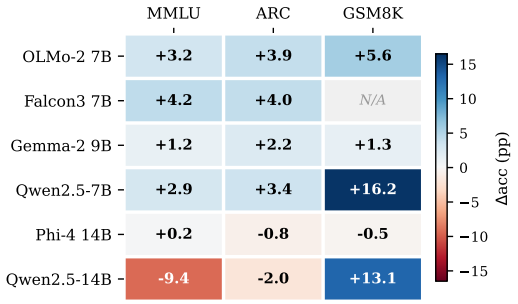


Figure 3: Cross-benchmark transfer at d100. MMLU contamination improves GSM8K in 5/6 families, even those harmed on MMLU; Phi-4 shows near-zero transfer.

target-benchmark immunity does not imply transfer immunity.

5.2 Cross-Benchmark Transfer

Contamination patterns are a property of the (model, benchmark) pair, not the model alone. Within MCQ format, MMLU–ARC direction agreement is 86% (6/7 families; Appendix B); across formats (MCQ→generative math), agreement drops to 40% (2/5) with three families exhibiting directional reversal on GSM8K (Cobbe et al., 2021) (Table 4, Appendix). All five non-excluded families show positive GSM8K transfer (+1 to +16pp), including Qwen-14B (−9.4pp MMLU yet +13.1pp GSM8K), suggesting capability elicitation rather than memorization, consistent with null DiD (§5.4). Volume effects are ruled out: d005 (+1.5% data) yields +11.1pp for Qwen-7B, while Phi-4 shows near-zero transfer.

5.3 Cross-Seed Stability

With $n = 3$ seeds per family (CI widths 1.5–14.5pp; Table 11, Appendix), 5/6 families receive consistent direction-level classifications under our aggregation rule (§3.4); direction is more reproducible than shape (strict sign consistency 3/6). OLMo-2's CI crosses zero, rendering its classification borderline.

5.4 Item-Level Memorization Test

We test whether dose-response patterns arise from item-specific memorization using the treated/untreated split at d025 (3,540 / 10,502 MMLU items). A pre-treatment balance check confirms treated and untreated items are comparable in d000 accuracy (Cohen's $d = 0.02$), difficulty distribution, and domain coverage (all $p > 0.19$), supporting the parallel trends assumption. No model shows significant DiD ($p > 0.05$; Table 2), ruling out item-specific effects >2pp (MDES 1.25–2.05pp; ARC MDES in Appendix P).

Table 2: Treated vs. untreated item accuracy changes at d025 MMLU (3,540 / 10,502). All DiD non-significant. $^{**}p < .01$; $^{***}p < .001$; ns: $p > .05$.

Model	Treated Δ	Untreated Δ	DiD	Verdict
Gemma-2 9B	$+3.1^{**}$	$+2.6^{***}$	+0.5 (ns)	generalization
OLMo-2 7B	+1.6 (ns)	$+2.7^{***}$	−1.1 (ns)	mixed
Phi-4 14.7B	+0.3 (ns)	+0.6 (ns)	−0.3 (ns)	no-effect
Qwen-14B	-6.9^{***}	-7.6^{***}	+0.7 (ns)	anti-contam.
Qwen-7B	-3.8^{**}	-2.9^{***}	−0.8 (ns)	anti-contam.

Parallel changes suggest global capability modulation (Figure 8, Appendix).

Volume control. A volume-matched control (Falcon3-7B trained on 66,533 non-benchmark MCQ items; Appendix Q) achieves 60.67% MMLU, decomposing the +4.19pp Benefit into data quantity (+1.90pp, 45%) and contamination content (+2.29pp, 55%); ARC yields a consistent 43%/57% split. Collapse (−9.4pp) and Immune (<1pp) cannot arise from volume effects that produce +1.90pp in the control.

5.5 Calibration Quality

Within-family IRT analysis uses Qwen2.5 ≤3B ($J = 16$), where aggregate contamination shifts accuracy by ∼0.6–0.7pp. RRF-guided item filtering achieves $\tau = 0.73$ on 300 items ($\Delta\tau = 0.18$ vs. size-matched random; signal ablation in Appendix E). IRT is fitted on these 300 highest-exposure-variance items (2.1% of MMLU; cutoff selected at the exposure-variance inflection point); contamination effects on remaining low-variance items are not empirically verified.

Table 3 compares methods on 300 RRF-filtered items. Raw accuracy's MAE advantage is predicted by Eq. (3): under near-null aggregate contamination, both estimators converge; ContamCal's value lies in the diagnostic pathway (γ_d). B6-DCR systematically overcorrects (MAE 0.108). Domain analysis reveals Social Sciences $\gamma_d = -0.63$ ($P(\gamma_d < 0) = 99.7\%$; §5.7), while pooled γ remains non-significant. Model fit diagnostics (PPP, CI coverage) are in Appendix K and X.

5.6 Detection Signal Failure Modes

Min-K%++ (Zhang et al., 2025) exhibits *directional reversal* at 0.5B: contaminated items re-

Table 3: MMLU calibration on 300 RRF-filtered items (Qwen2.5, J=16). ARC results in Appendix X.

Method	MAE	τ
B0: Raw Accuracy	**0.006**	0.767
B1b: Remove & Rescore	0.026	<u>0.783</u>
B3: Vanilla 2PL	0.008	0.731
B4: Linear Regression	0.010	0.751
B5b: Emperor	**0.006**	**0.789**
B6: DCR (reimpl.)	0.108	0.654
ContamCal (2PL+γ)	0.009	0.731

ceive *lower* scores (paired Cohen's $d = -1.98$ at d100, $|d| > 1.4$ at d005). S1 achieves AUC > 0.95 at all scales; S4 is near-random below 1.5B (Appendix D). Simulation shows γ recovery requires AUC$\geq$0.55 with $J \geq 50$; at AUC ≥ 0.90, $J = 16$ suffices (Appendix F).

5.7 Domain-Level Analysis

Within-family domain analysis (Qwen2.5, $J = 16$) reveals heterogeneity masked by aggregate scores: STEM domains show positive γ_d while Social Sciences is negative ($\gamma_d = -0.63$, 95% CI $[-1.07, -0.19]$, $P(\gamma_d < 0) = 99.7\%$, PPP=0.722; Figure 11, Table 16 in Appendix). Social Sciences $P(\gamma_d < 0)$ is 99.35–99.85% across priors (Table 9), exceeding the Bonferroni threshold (99.375%) under baseline and loose priors but falling 0.03 pp short under the tight prior. The directional signal is robust; we treat domain-level findings as exploratory pending replication with larger ensembles.

6 Discussion and Conclusion

Controlled SFT injection reveals a spectrum of dose-response patterns—not uniform inflation. Benefit and Immune replicate across independent families; Collapse and V-recovery (65–70% baseline band) are each attested in one family (Qwen2.5) and await replication. DiD finds no memorization (MDES 1.2–2.0pp); direction reproduces across seeds but magnitude is seed-sensitive. Out-of-sample validation (LLaMA-3.1 Immune despite matching OLMo-2's Benefit baseline) confirms model family characteristics beyond baseline ability modulate vulnerability. We hypothesize that LLaMA-3.1's $\sim$3$\times$ larger pre-training corpus (15T vs. 5T tokens) produces representations more resistant to SFT perturbation (Bordt et al., 2025b), while OLMo-2's two-phase annealing and Dolma-based composition retain greater plasticity—consistent with its pronounced cross-seed sensitivity.

Implications for practice. Multi-seed dose-response profiling at $\geq$3 dosages can detect vulnerability in the 65–70% baseline band; domain γ_d heterogeneity (up to 1.8$\times$) motivates stronger contamination controls, and ContamCal can flag anomalous profiles given $J \geq 50$.

Limitations

All experiments use models at or below 14.7B parameters; 70B+ models may exhibit qualitatively different contamination dynamics where SFT injection is absorbed without measurable signal divergence. With $n = 3$ seeds per family, effect *direction* is reproducible but magnitude is seed-sensitive (OLMo-2's cross-seed CI crosses zero, rendering its Benefit classification borderline), and borderline classifications should be interpreted with caution. Finally, our LoRA-based SFT injection in AUGMENT mode differs from pretraining-stage leakage via web crawl ingestion, and our dosage levels assume uniform injection of all benchmark items; real-world contamination may follow skewed exposure distributions, potentially producing different dose-response profiles. Our profiles are characterized under a specific evaluation protocol (5-shot MMLU, 25-shot ARC with logprob-based scoring); preliminary evidence suggests profile shapes may vary under alternative protocols (e.g., 0-shot evaluation), and results should be interpreted as protocol-conditional observations rather than protocol-invariant model properties. Unified LoRA rank ($r = 16$) yields a parameter fraction inversely proportional to model size—specifically, 0.45% for Phi-4 (14.7B) and 0.59% for Gemma-2 (9B); Phi-4's immunity may partly reflect insufficient adaptation capacity, though Qwen-14B's collapse at the same scale argues against a purely capacity-based explanation. Out-of-sample validation covers only the Immune pattern; independent-family replication of Collapse and V-recovery remains open (both are Qwen2.5-specific in our sample). Benefit is replicated in three independent families (OLMo-2, Falcon3, Gemma-2). All single-family patterns are, however, replicated across three random seeds. The AUGMENT protocol increases total training volume with dosage (+33% at d100); a volume-matched control attributes approximately 45% of Falcon3's Benefit to data quantity (§5.4), though contamination

content (55%) remains the majority contributor. IRT is fitted on the 300 highest-exposure-variance items (2.1% of MMLU); contamination effects on the remaining low-variance items may differ. The 5%→25% dosage gap may miss inflection points in non-monotonic patterns. Our findings are primarily driven by MMLU (14,042 items); ARC-Challenge (1,172 items) provides limited statistical power (MDES 2.3–5.6pp), and GSM8K results are secondary. Our design cannot isolate architecture from co-varying factors (e.g., pre-training data volume, curriculum); our SFT-based injection complements pretraining-stage studies (Schaeffer et al., 2026).

Ethics Statement

All experiments use publicly available, open-weight language models (Qwen2.5, Phi-4, OLMo-2, Gemma-2, Falcon3, and LLaMA-3.1) under their respective licenses. Contamination injection is performed via standard supervised fine-tuning on existing public benchmark items; no new data collection involving human subjects is conducted. All evaluation benchmarks (MMLU, ARC-Challenge, GSM8K) are publicly released datasets. This work does not involve private user data, personally identifiable information, or human annotation.

We acknowledge a dual-use risk: the contamination injection protocol we describe could be used to deliberately inflate benchmark scores. We mitigate this by focusing on detection and measurement rather than injection optimization, and by releasing detection signals rather than injection recipes. Our findings characterize the conditions under which SFT injection affects aggregate scores, which may inform decisions about manipulation risk at different scales. The measurement framework is intended to improve evaluation transparency, not to enable gaming.

Code for ContamCal, all detection signals, fine-tuned checkpoints, and evaluation scripts will be released upon publication at `[anonymized]`.

References

Marah Abdin, Sébastien Bubeck, Ronen Eldan, Suriya Gunasekar, Neel Jain, Yin Tat Lee, Yuanzhi Li, Arindam Saha, Adil Salim, and 1 others. 2024. Phi-4 technical report. *arXiv preprint arXiv:2412.08905*.

Simone Balloccu, Patrícia Schmidtová, Mateusz Lango, and Dominik Dusička. 2024. Leak, cheat, repeat: Data contamination and evaluation malpractices in closed-source LLMs. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*.

Sebastian Bordt, Harsha Nori, and Rich Caruana. 2025a. How much can we forget about data contamination? *arXiv preprint*.

Sebastian Bordt, Suraj Srinivas, Valentyn Boreiko, and Ulrike von Luxburg. 2025b. How much can we forget about data contamination? In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*.

Gregory Camilli and Lorrie A. Shepard. 1994. *Methods for Identifying Biased Test Items*. Measurement Methods for the Social Sciences. Sage Publications, Thousand Oaks, CA.

Nicholas Carlini, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Florian Tramèr, and Chiyuan Zhang. 2023. Quantifying memorization across neural language models. In *Proceedings of the 11th International Conference on Learning Representations (ICLR)*.

Bob Carpenter, Andrew Gelman, Matthew D. Hoffman, Daniel Lee, Ben Goodrich, Michael Betancourt, Marcus Brubaker, Jiqiang Guo, Peter Li, and Allen Riddell. 2017. Stan: A probabilistic programming language. *Journal of Statistical Software*, 76(1):1–32.

Yupeng Chang, Xu Wang, Jindong Wang, Yuan Wu, Linyi Yang, Kaijie Zhu, Hao Chen, Xiaoyuan Yi, Cunxiang Wang, Yidong Wang, Wei Ye, Yue Zhang, Yi Chang, Philip S. Yu, Qiang Yang, and Xing Xie. 2024. A survey on evaluation of large language models. *ACM Transactions on Intelligent Systems and Technology*, 15(3):1–45.

Simin Chen, Yiming Chen, Zexin Li, Yifan Jiang, Zhongwei Wan, Yixin He, Dezhi Ran, Tianle Gu, Haizhou Li, Tao Xie, and Baishakhi Ray. 2025. Benchmarking large language models under data contamination: A survey from static to dynamic evaluation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 10080–10098.

Hyeong Kyu Choi, Maxim Khanov, Hongxin Wei, and Yixuan Li. 2025. How contaminated is your benchmark? measuring dataset leakage in large language models with kernel divergence. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*.

Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. Think you have solved question answering? Try ARC, the AI2 reasoning challenge. *arXiv preprint*.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. Training verifiers to solve math word problems. *arXiv preprint*.

Gordon V. Cormack, Charles L. A. Clarke, and Stefan Buettcher. 2009. Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 758–759. ACM.

R. J. De Ayala. 2009. *The Theory and Practice of Item Response Theory*. Guilford Press, New York.

Paul De Boeck and Mark Wilson. 2004. *Explanatory Item Response Models: A Generalized Linear and Nonlinear Approach*. Springer, New York.

Jasper Dekoninck, Mark Niklas Müller, and Martin Vechev. 2024. ConStat: Performance-based contamination detection in large language models. In *Advances in Neural Information Processing Systems 37 (NeurIPS)*.

Yihong Dong, Xue Jiang, Huanyu Liu, Zhi Jin, Bin Gu, Mengfei Yang, and Ge Li. 2024. Generalization or memorization: Data contamination and trustworthy evaluation for large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*.

Carol A. Eckerly. 2017. Detecting preknowledge and item compromise: Understanding the status quo. In Gregory J. Cizek and James A. Wollack, editors, *Handbook of Quantitative Methods for Detecting Cheating on Tests*, chapter 5. Routledge.

Andrew Gelman, Jennifer Hill, and Masanao Yajima. 2012. Why we (usually) don't have to worry about multiple comparisons. *Journal of Research on Educational Effectiveness*, 5(2):189–211.

Gemma Team. 2024. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*.

Shahriar Golchin and Mihai Surdeanu. 2024. Time travel in LLMs: Tracing data contamination in large language models. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, and 1 others. 2024. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*.

Yuling Gu, Oyvind Tafjord, Bailey Kuehl, Dany Haddad, Jesse Dodge, and Hannaneh Hajishirzi. 2024. OLMES: A standard for language model evaluations. *arXiv preprint*.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. 2017. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, pages 1321–1330.

Eliya Habba, Itay Itzhak, Asaf Yehudai, Yotam Perlitz, Elron Bandel, Michal Shmueli-Scheuer, Leshem Choshen, and Gabriel Stanovsky. 2026. Growing pains: Extensible and efficient LLM benchmarking via fixed parameter calibration. *arXiv preprint*.

Ronald K. Hambleton, Hariharan Swaminathan, and H. Jane Rogers. 1991. *Fundamentals of Item Response Theory*. Sage Publications, Newbury Park, CA.

Jongwook Han, Woojung Song, Jonggeun Lee, and Yohan Jo. 2026. Quantifying data contamination in psychometric evaluations of LLMs. In *Findings of the Association for Computational Linguistics: EACL 2026*, pages 6070–6088.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2021. Measuring massive multitask language understanding. In *Proceedings of the 9th International Conference on Learning Representations (ICLR)*.

Archibald Vivian Hill. 1910. The possible effects of the aggregation of the molecules of haemoglobin on its dissociation curves. *Journal of Physiology*, 40:iv–vii.

Matthew D. Hoffman and Andrew Gelman. 2014. The No-U-Turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15(47):1593–1623.

Paul W. Holland and Howard Wainer. 1993. *Differential Item Functioning*. Lawrence Erlbaum Associates, Hillsdale, NJ.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, Scott Johnston, Sheer El-Showk, Andy Jones, Nelson Elhage, Tristan Hume, Anna Chen, Yuntao Bai, Sam Bowman, Stanislav Fort, and 17 others. 2022. Language models (mostly) know what they know. *arXiv preprint*.

Murat Yildirim Kocyigit and Ekin Akyurek Yildirim. 2025. The impact of post-training on data contamination detection in large language models. In *Advances in Neural Information Processing Systems 38 (NeurIPS)*.

John P. Lalor, Pedro Rodriguez, João Sedoc, and Jose Hernandez-Orallo. 2024. Item response theory for natural language processing. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: Tutorial Abstracts (EACL)*.

Changmao Li and Jeffrey Flanigan. 2024. Task contamination: Language models may not be few-shot anymore. In *Proceedings of the 38th AAAI Conference on Artificial Intelligence*, volume 38, pages 18471–18480.

Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, Benjamin Newman, Binhang Yuan, Bobby Yan, Ce Zhang, Christian Cosgrove, Christopher D. Manning, Christopher Ré, Diana Acosta-Navas, Drew A. Hudson, and 31 others. 2023. Holistic evaluation of language models. *Annals of the New York Academy of Sciences*, 1525(1):140–146.

Frederic M. Lord. 1980. *Applications of Item Response Theory to Practical Testing Problems*. Lawrence Erlbaum Associates, Hillsdale, NJ.

Lennart Luettgau, Harry Coppock, Magda Dubois, Christopher Summerfield, and Cozmin Ududec. 2025. HiBayES: A hierarchical Bayesian modeling framework for AI evaluation statistics. *arXiv preprint*.

Inbal Magar and Roy Schwartz. 2022. Data contamination: From memorization to exploitation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 157–165, Dublin, Ireland. Association for Computational Linguistics.

Yonatan Oren, Nicole Meister, Niladri Chatterji, Faisal Ladhak, and Tatsunori Hashimoto. 2024. Proving test set contamination in black box language models. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*.

Felipe Maia Polo, Lucas Weber, Leshem Choshen, Yuekai Sun, Gongjun Xu, and Mikhail Yurochkin. 2024. tinyBenchmarks: Evaluating LLMs with fewer examples. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*.

Christian Ritz, Florent Baty, Jens C. Streibig, and Daniel Gerhard. 2015. *Dose-Response Analysis Using R*. CRC Press.

Pedro Rodriguez, Joe Barrow, Alexander Miserlis Hoyle, John P. Lalor, Robin Jia, and Jordan Boyd-Graber. 2021. Evaluation examples are not equally informative: How should that change NLP leaderboards? In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 4486–4503.

Oscar Sainz, Jon Ander Campos, Iker García-Ferrero, Julen Etxaniz, Oier Lopez de Lacalle, and Eneko Agirre. 2023. NLP evaluation in trouble: On the need to measure LLM data contamination for each benchmark. In *Findings of the Association for Computational Linguistics: EMNLP 2023*.

Ernesto San Martín, Guido Del Pino, and Paul De Boeck. 2009. Consequential validity and item response theory. *Psychometrika*, 74(2):349–367.

Rylan Schaeffer, Joshua Kazdan, Baber Abbasi, Ken Ziyu Liu, Brando Miranda, Ahmed Ahmed, Fazl Berez, Abhay Puri, Stella Biderman, Niloofar Mireshghallah, and Sanmi Koyejo. 2026. Quantifying the effect of test set contamination on generative evaluations. *arXiv preprint*.

Ulrich Schroeders and Timo Gnambs. 2025. Sample-size planning in item-response theory: A tutorial. *Advances in Methods and Practices in Psychological Science*.

Weijia Shi, Anirudh Ajith, Mengzhou Xia, Yangsibo Huang, Daogao Liu, Terra Blevins, Danqi Chen, and Luke Zettlemoyer. 2024. Detecting pretraining data from large language models. In *Proceedings of the 12th International Conference on Learning Representations (ICLR)*.

Aaditya K. Singh, Muhammed Yusuf Kocyigit, Andrew Poulton, David Esiobu, Maria Lomeli, Gergely Szilvasy, and Dieuwke Hupkes. 2024. Evaluation data contamination in LLMs: How do we measure it and (when) does it matter? *arXiv preprint*.

Sandip Sinharay. 2017. Detection of item preknowledge using likelihood ratio test and score test. *Journal of Educational and Behavioral Statistics*, 42(1).

Yifan Sun, Han Wang, Dongbai Li, Gang Wang, and Huan Zhang. 2025. The emperor's new clothes: Systematic evaluation of contamination mitigation strategies for large language models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*. ArXiv:2503.16402.

Hariharan Swaminathan and H. Jane Rogers. 1990. Detecting differential item functioning using logistic regression procedures. *Journal of Educational Measurement*, 27(4):361–370.

Yongding Tao, Tian Wang, Yihong Dong, Huanyu Liu, Kechi Zhang, Xiaolong Hu, and Ge Li. 2025. Detecting data contamination from reinforcement learning post-training for large language models. *arXiv preprint*.

Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groeneveld, Kyle Lo, and 1 others. 2025. OLMo 2: The best fully open language model to date. *arXiv preprint arXiv:2501.00656*.

Technology Innovation Institute. 2024. The Falcon 3 family of open models.

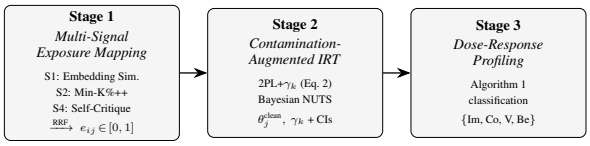


Figure 4: ContamCal three-stage pipeline. Stage 1 fuses training-free detection signals into per-item exposure scores via Reciprocal Rank Fusion. Stage 2 embeds these scores in a contamination-augmented 2PL IRT model, recovering clean ability estimates with credible intervals. Stage 3 classifies the resulting dose-response trajectories into canonical patterns.

Colin White, Samuel Dooley, Manley Roberts, Arka Pal, Benjamin Feuer, Siddhartha Jain, Ravid Shwartz-Ziv, Neel Joshi, and Micah Goldblum. 2024. LiveBench: A challenging, contamination-free LLM benchmark. *arXiv preprint*.

Cheng Xu, Nan Yan, Shuhao Guan, Changhong Jin, Yuke Mei, Yibing Guo, and Tahar Kechadi. 2025. DCR: Quantifying data contamination in LLMs evaluation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 23002–23020.

Ruijie Xu, Zengzhi Wang, Run-Ze Fan, and Pengfei Liu. 2024. Benchmarking benchmark leakage in large language models. *arXiv preprint*.

An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Junyang Lin, and Jingren Zhou. 2024. Qwen2.5 technical report. *arXiv preprint*.

Shuo Yang, Wei Wei, Ganqu Liu, Zhengxuan Liu, and Bowen Yu. 2023. Rethinking benchmark and contamination for language models with rephrased samples. *arXiv preprint*.

Hugh Mee Zhang, Jeff Da, Nayeon Lee, Lianmin Zheng, Ying Sheng, Joseph E. Gonzalez, and Ion Stoica. 2024. A careful examination of large language model performance on grade school arithmetic. In *arXiv preprint*.

Jingyang Zhang, Jingwei Sun, Eric Yeats, Yang Ouyang, Martin Kuo, Jianyi Zhang, Hao Yang, and Hai Li. 2025. Min-K%++: Improved baseline for detecting pre-training data from large language models. In *Proceedings of the 13th International Conference on Learning Representations (ICLR)*.

Hongli Zhou, Hainiu Wang, Qingyao Ye, and Nanyun Peng. 2026. Lost in benchmarks? rethinking large language model benchmarking with item response theory. In *Proceedings of the 40th AAAI Conference on Artificial Intelligence*.

Table 4: Cross-benchmark transfer: MMLU vs. GSM8K accuracy change at d100. LLaMA-3.1 shows the strongest asymmetric transfer (+16.4pp GSM8K despite MMLU immunity). Gemma-2 excluded due to CoT collapse under MC-format SFT.

Model	MMLU Δ	GSM8K Δ	Range	Transfer
Phi-4 14.7B	±0.5pp	−0.5pp	—	Null
OLMo-2 7B	+3.2pp	+2.1pp[§]	—	Consistent
Qwen-7B	−1.8pp	+11.1pp	±0.9	Reversal
Qwen-14B	−9.4pp	+13.1pp	±7.4	Reversal[†]
LLaMA-3.1 8B	−0.7pp	+16.4pp[‡]	—	Asymmetric

[†] d000 baseline seed-sensitive (s42=60.8% vs. s123=76.7%); d100 stable (82 ± 0.5%).
[§] Mean across seeds: +2.1pp (s42: +1.7pp, s123: +2.4pp); s0 shows +9.2pp.
[‡] Single seed (s0); MMLU Immune but strongest GSM8K transfer across all families.

A Cross-Benchmark Transfer Details

B ARC-Challenge Cross-Benchmark Analysis

ARC-Challenge (25-shot, 1,172 items) patterns under s42 are directionally consistent with MMLU for five of six models: OLMo-2 benefits (+3.9pp), Falcon3 benefits (+4.0pp), Qwen-7B (−2.9pp) and Qwen-14B (−3.4pp) are harmed, Phi-4 remains immune (+0.6pp). Gemma-2 achieves $n = 3$ on ARC: s42 exhibits a 16pp oscillation artifact (endpoint −1.20pp), s0 shows harm (−1.97pp), and s123 shows benefit (+2.73pp). The $n = 3$ ARC mean is −0.15pp (SD=2.41pp; BCa 95% CI [−1.97, +1.42]), with cross-seed direction inconsistent (−/−/+). This contrasts with MMLU, where all three seeds show benefit (mean +2.38pp, CI [+1.49, +3.14] excludes zero), making Gemma-2 the clearest example of benchmark-conditional contamination effects. Bootstrap stability is substantially lower than MMLU (OLMo-2: 53.3%, Gemma-2: 62.7%) due to the smaller item pool (1,172 vs. 14,042).

Cross-seed ARC consistency. Phi-4 is the strongest anchor with $n = 3$ on both benchmarks: ARC mean Δ = +0.66pp (SD=0.18pp; s0 +0.51, s42 +0.60, s123 +0.86; BCa 95% CI [+0.51, +0.77]), MMLU mean Δ = +0.73pp—cross-benchmark agreement within 0.07pp. While statistically detectable on ARC (CI excludes zero, unlike MMLU CI [−0.17, +1.39]), the effect (+0.66pp) is too small to meaningfully inflate benchmark scores, consistent with the Immune classification. Qwen-7B achieves $n = 3$ on ARC: mean Δ = −2.42pp (SD=0.51pp; s0 −1.88, s42 −2.90, s123 −2.47; BCa 95% CI [−2.90, −2.08]), confirming persistent harm. The d005 V-dip severity is seed-dependent (−1.8pp

vs. −4.7pp). Qwen-14B achieves $n = 3$ on both benchmarks: MMLU mean $\Delta = -15.7$pp (BCa CI [−24.99, −10.49]; s0 most extreme at −25.0pp due to position bias amplification); ARC mean $\Delta = -5.03$pp (BCa CI [−9.38, −2.76]). Both CIs exclude zero, confirming Collapse. OLMo-2 achieves $n = 3$ on ARC: mean $\Delta = +3.44$pp (SD=0.52pp; s0 +3.50, s42 +3.92, s123 +2.90). Despite being the most seed-sensitive family on MMLU (SD=1.9pp), OLMo-2 is remarkably stable on ARC (SD=0.52pp)—seed sensitivity is itself benchmark-dependent, likely modulated by item count and difficulty distribution. Using multi-seed MMLU means (+1.4pp), direction is consistently positive on both benchmarks, illustrating why cross-benchmark comparisons require multi-seed aggregation (§5.3). ARC dose-response trajectories (s123): Qwen-14B 90.4→91.5→89.8→88.7→87.3 (d005 transient spike then monotonic decline, consistent with MMLU s123 shape); Qwen-7B 87.1→82.4→85.2→85.1→84.7 (d005 sharp drop then partial recovery); OLMo-2 76.9→77.2→76.6→79.2→79.8 (monotonic rise).

Cross-seed direction on ARC is consistent for 4/5 families: Phi-4 and OLMo-2 positive across three seeds each, Qwen-7B and Qwen-14B negative across three seeds each. Gemma-2 is the exception (−/−/+ across seeds; mean −0.15pp). This contrasts with MMLU, where only 2/5 families achieve strict sign consistency (§5.3). ARC yields tighter confidence intervals and higher cross-seed consistency than MMLU, likely because its smaller item pool (1,172 vs. 14,042) amplifies the contamination signal—suggesting that benchmark size modulates not only effect magnitude but also statistical detectability.

Cross-benchmark direction (MMLU vs. ARC) is consistent for 4/6 families (§5.2): Qwen-14B (Harm), Phi-4 (Immune), OLMo-2 (Benefit), and Falcon3 (Benefit). Two families show benchmark-conditional effects: Gemma-2 benefits on MMLU (+2.38pp, CI excludes zero) but shows no significant ARC effect (−0.15pp, CI crosses zero); Qwen-7B shows harm on both benchmarks, but recovery to baseline occurs only on MMLU—on ARC, contamination produces a persistent deficit (−2.4pp mean across three seeds). Effect magnitudes remain benchmark-specific: Qwen-14B suffers −5.0pp on ARC compared to −15.7pp on MMLU (3.1× attenuation), implying that while direction transfers, magnitude does not. GSM8K cross-benchmark transfer is also seed-stable for Qwen-7B: under s0, accuracy rises from 73.3% to 81.2% (+7.9pp, monotonic), consistent with the directional reversal observed under s42 (+11.1pp; Table 4).

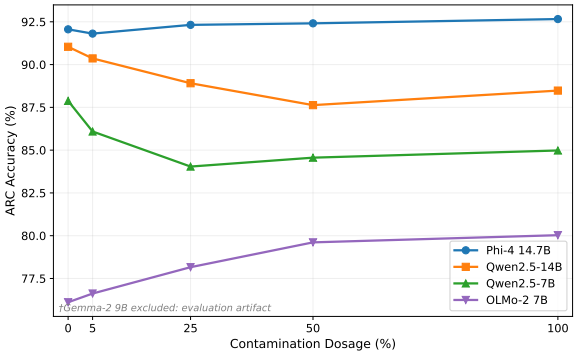


Figure 5: ARC-Challenge accuracy vs. contamination dosage (25-shot). Patterns are directionally consistent with MMLU (Figure 1) for 4/5 models.

C GSM8K Dose-Response Transfer

Table 5 reports GSM8K accuracy across contamination dosages for all six families (single seed, s0). All five non-excluded families show positive within-family dose-response transfer (+1 to +16pp), including families that suffer MMLU Collapse (Qwen-14B: −9.4pp MMLU, +7.4pp GSM8K) or are MMLU-Immune (LLaMA-3.1: −0.7pp MMLU, +16.4pp GSM8K—the strongest transfer effect). This directional reversal—MMLU-targeted contamination improving an unrelated open-ended math benchmark—strengthens the global capability modulation interpretation. Within-family dose-response patterns (direction and relative magnitude of Δ) are the primary finding, as all dosage levels for each family use identical evaluation settings.

Table 5: GSM8K accuracy (%) across MMLU contamination dosages (seed=0). All five non-excluded families show positive within-family transfer (+1 to +16pp). Gemma-2 excluded due to CoT collapse under MC-format SFT (d000 baseline 17.8%).

Family	d000	d005	d025	d050	d100	Δ
Phi-4 14.7B	84.46	—	—	—	85.60	+1.14
Qwen2.5-14B	74.98	71.19	79.30	81.05	82.41	+7.43
Qwen2.5-7B	73.31	78.32	78.24	79.38	81.20	+7.89
OLMo-2 7B	60.58	—	—	—	69.75	+9.17‡
LLaMA-3.1 8B	38.74	—	52.16	—	55.12	+16.38
Gemma-2 9B	17.82†	—	—	—	—	—

†CoT collapse under MC-format SFT; excluded from analysis.
‡Seed-sensitive: s0 +9.17pp vs. s42 +2.1pp (Table 4).

GSM8K evaluations use

max_new_tokens=256, which may truncate chain-of-thought reasoning for some models. Absolute accuracy is therefore a lower bound; cross-family comparisons of absolute values should be interpreted with caution. Within-family dose-response patterns (direction and relative magnitude of Δ) are the primary finding, as all dosage levels for each family use identical evaluation settings.

D Signal Reliability Across Scales

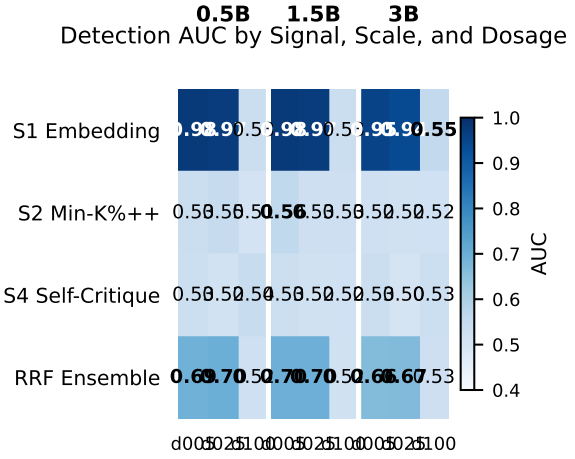


Figure 6: Detection signal AUC across scale, dosage, and signal type. S1 achieves AUC > 0.95 at all scales; S2 (Min-K%++) reverses at 0.5B; S4 is near-random below 1.5B.

E RRF Signal Ablation

Signal orthogonality. Table 6 reports pairwise Spearman rank correlations between detection signals at the item level, averaged across 31 models. All pairwise $|\rho| < 0.06$, confirming that S1, S2, and S4 capture orthogonal aspects of contamination exposure.

Table 6: Pairwise Spearman ρ between detection signals (item-level, averaged across 31 models). All $|\rho| < 0.06$.

Signal Pair	MMLU	ARC	GSM8K
S1–S2	−0.044	−0.019	−0.022
S1–S4	+0.038	+0.009	+0.053
S2–S4	−0.001	+0.012	+0.007

Leave-one-out ablation. Table 7 reports the effect of removing each signal from the RRF ensemble. ρ_{dose} is the model-level Spearman correlation between mean exposure score and true dosage; ΔMAE is the change in model-level exposure–dosage MAE vs. the full configuration; item ρ is the rank correlation of item-level exposure scores vs. the full configuration.

Table 7: Leave-one-signal-out RRF ablation (31 models, $\kappa = 60$).

Configuration	ρ_{dose}	ΔMAE	Item ρ
MMLU			
Full (S1+S2+S4)	0.049	—	—
Drop S1	0.045	+0.066	0.782
Drop S2	**0.471**	+0.045	0.773
Drop S4	0.084	+0.079	0.786
ARC-Challenge			
Full (S1+S2+S4)	−0.003	—	—
Drop S1	−0.006	−0.054	0.784
Drop S2	0.029	−0.079	0.786
Drop S4	0.001	−0.004	0.784

Dropping S2 (Min-K%++) causes ρ_{dose} to jump from 0.049 to 0.471 on MMLU, revealing that S2 actively suppresses model-level dosage discrimination. This is consistent with the directional reversal at small scales (§5.6): S2 assigns higher scores to clean items in some models, counteracting S1 and S4. Despite this model-level harm, S2 provides unique item-level information (item $\rho = 0.773$, i.e., $\sim$23% of item rankings change upon removal). Signal importance by ΔMAE ranks S4 $>$ S1 $>$ S2 on MMLU; this ordering holds across benchmarks. The near-zero ρ_{dose} (0.049 on MMLU, -0.003 on ARC) confirms that RRF exposure is an item-level tool: it identifies which items within a model are most likely contaminated, but cannot distinguish dosage levels across models.

Signal contribution to γ estimation. Leave-one-out ablation on γ recovery (Qwen2.5, $J = 16$) confirms S4 (self-critique) as the dominant contributor: removing S4 incurs $4.9\times$ the γ estimation loss of removing S2 (Min-K%++). No signal is pure noise: the worst single-signal $\rho(E, \tau)$ is 0.072–0.118 (S2), still $5.5\times$ the random baseline. Marginal returns from three to two signals are negligible (dropping S2 leaves ρ effectively unchanged), suggesting the current three-signal ensemble is near-optimal for the Qwen2.5 family.

F Identifiability Simulation

Fisher information $\mathcal{I}(\gamma) \propto \text{Var}(e_{\cdot i})$ collapses when signal AUC is low. At AUC=0.55, bias < 0.05 and 95% CI coverage $\geq 90\%$ require

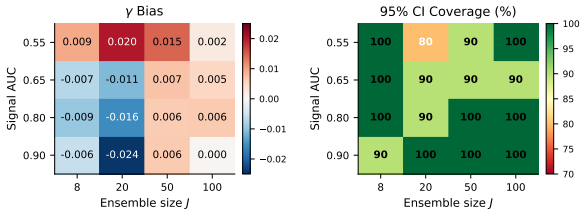


Figure 7: γ recovery as a function of signal AUC and ensemble size J. Bias < 0.05 and 95% CI coverage $\geq 90\%$ require $J \geq 50$ at AUC=0.55, but only $J \geq 16$ at AUC$\geq$0.90.

$J \geq 50$; at AUC$\geq$0.90 (as S1 achieves in practice), $J = 16$ suffices. CI width scales as $J^{-0.48}$, above current practice ($J = 8$–30; Habba et al., 2026; Lalor et al., 2024; Schroeders and Gnambs, 2025).

G 4PL Extension

For MC-format benchmarks where guessing and ceiling effects are non-negligible, we extend to:

$$P(y_{ijk} = 1) = c_i + (d_i - c_i)\,\sigma\big(a_i(\theta_j - b_i) + \gamma_k \cdot e_{ij}\big), \quad (4)$$

with $c_i \sim \text{Beta}(5, 20)$ and $d_i \sim \text{Beta}(20, 5)$. This requires $J \geq 16$ to avoid overparameterization; at $J = 5$, 4PL+γ yields PPP=0.000 and MAE degradation from 0.004 to 0.068. Our domain-level analyses use 2PL+γ at $J = 13$; pooled calibration uses $J = 16$.

H S4 Self-Critique Signal Details

S4 prompts the model with "Is the following answer correct?" followed by the question and candidate answer. We extract YES/NO logits to compute $P(\text{YES})$, then score each item as the negative entropy of this distribution: $\text{S4}_i = -[-p \log p - (1-p)\log(1-p)]$ where $p = P(\text{YES})$. Scores near zero indicate high confidence (potentially contaminated); large negative values indicate uncertainty.

I Training Convergence

All SFT configurations converged. Final training loss increases monotonically with dosage (d000: 1.08, d005: 1.08, d025: 1.14, d100: 1.17 averaged across seeds), consistent with distribution shift from benchmark item injection.

J Model Exclusion Details

Of the 16 models remaining after 0.5B exclusion, three 3B checkpoints were removed: two had mismatched naming conventions between evaluation and signal extraction stages, and one lacked evaluation results under the unified hyperparameter setting ($\alpha = 32$). Additionally, 3B d000 (clean control) checkpoints were excluded because the standardized protocol (LoRA $\alpha = 32$, lr$= 5 \times 10^{-5}$) was applied only to dosages d005–d100 at the 3B scale; earlier d000 checkpoints used inconsistent hyperparameters across retraining iterations.

Model sets across analyses. Table 8 summarizes model compositions. Model sets vary because domain-level runs preceded the 3B retraining (§4.1). Core findings (γ non-significance, raw accuracy dominance) replicate across $J = 13$ and $J = 16$ configurations.

Table 8: Model set composition across analyses.

Analysis	J	Model Set	Section
Pooled baselines	16	8×Qwen-1.5B + 8×Qwen-3B	§5.5
Domain γ	13	8×Qwen-1.5B + 5×Qwen-3B	§5.7
Cross-family	16–28	varies by benchmark	§5.1
Oracle ablation	13	domain whitelist	§5.5

K Prior Sensitivity Analysis

To assess the influence of the γ prior, we refit the pooled MMLU model ($J = 16$, 300 RRF items) under three specifications: $\gamma \sim \mathcal{N}(0, 0.5)$, $\mathcal{N}(0, 1)$ (default), and $\mathcal{N}(0, 2)$. The posterior mean shifts from 0.34 to 0.89 across priors, while the 95% CI includes zero in all cases (lower bound stable at ≈ -0.4; upper bound ranges from 1.11 to 2.20). All fits converge ($\hat{R} < 1.01$, ESS > 3800, PPP $\in [0.16, 0.25]$). The prior sensitivity confirms that $J = 16$ provides insufficient data to override the prior, which reinforces the exploratory status of γ-based conclusions at current ensemble sizes.

For the Social Sciences domain (corrected $J = 16$ pool, 2000 items), the directional signal is robust to prior choice:

Table 9: Social Sciences γ prior sensitivity (corrected J=16 pool).

Prior	$\hat{\gamma}$	$P(\gamma < 0)$
$\mathcal{N}(0, 0.5)$	-0.541	99.35%
$\mathcal{N}(0, 1)$	-0.627	99.7%
$\mathcal{N}(0, 2)$	-0.654	99.85%

Social Sciences row updated with corrected $J = 16$ model pool; other domains retain the original prior sensitivity pool.

In contrast to the pooled analysis where $\hat{\gamma}$ shifts 2.6× across priors, Social Sciences $\hat{\gamma}$ varies only

1.2× (−0.54 to −0.65), with $P(\gamma < 0) \geq 99.35\%$ in all cases. The directional signal is robust to prior choice, though the tight prior falls marginally below the Bonferroni threshold (99.375% for eight domains).

L IRT Ability Separation by Scale

Table 10 reports the mean posterior θ estimates for 1.5B and 3B model groups across all MMLU domains and the pooled analysis. The ability gap $\Delta\theta = \bar{\theta}_{3B} - \bar{\theta}_{1.5B}$ is positive and significant (95% CI excludes zero) for all nine comparisons, which confirms that the IRT model recovers the expected scale-ability ordering.

Table 10: IRT ability estimates by model scale (J = 13 Qwen2.5). $\Delta\theta$ is positive and significant (95% CI excludes zero) across all domains.

Domain	$\bar{\theta}_{1.5B}$	$\bar{\theta}_{3B}$	$\Delta\theta$ [95% CI]
Math	−0.532	0.119	0.651 [0.570, 0.731]
Science	0.321	0.780	0.459 [0.370, 0.546]
CS/Eng	0.372	0.542	0.169 [0.030, 0.304]
Medical	0.521	1.053	0.531 [0.452, 0.609]
Humanities	0.228	0.496	0.268 [0.202, 0.337]
Social Sci	0.940	1.540	0.599 [0.523, 0.675]
Business/Law	0.083	0.417	0.334 [0.261, 0.410]
Other	0.817	1.565	0.748 [0.647, 0.846]
Pooled	0.311	0.811	0.500 [0.431, 0.568]

M Cross-Seed Stability Details

Table 11: Cross-seed Δacc (d000→d100). Strict sign consistency (Cons.) is 3/6; classification-level direction agreement is 5/6 (see text). Per-seed shape classification‡ agrees for only 3/6.

Model	n	s42	s123	s0	95% CI	Dir.	Cons.	Pattern‡
Phi-4	3	0	+	0	[−0.17, +1.39][B]	Immune	×	—
Qwen-14B	3	−	−	−	[−24.99, −10.49][B]	Harm	✓	Collapse
Qwen-7B	3	−	0	−	[−1.82, −0.34][B]	Harm	×	V-recovery
Gemma-2	3	+	+	+	[+1.49, +3.14][B]	Benefit	✓	Benefit
OLMo-2	3	+	0	+	[−0.55, +2.60][B]	Borderl.	×	—[b]
Falcon3-7B	2	+	+	—	[+3.91, +4.19][c]	Benefit	✓	Benefit

[B] BCa bootstrap ($B = 10{,}000$); original five families at $n = 3$.
‡ Per-seed shape may differ from family-level direction-based classification (§3.4).
[b] +/0/+ across seeds. Signs: +/− = |Δ|≥1pp; 0 = |Δ|<1pp.
[c] $n = 2$ seeds only; range shown instead of BCa CI.

OLMo-2 s42 rerun (d025/d100) exhibits ±1.6 pp point-estimate variability attributable to non-deterministic CUDA operations; this falls within the cross-seed range (3.76 pp) and does not affect pattern classification.

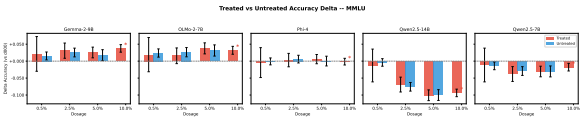


Figure 8: DiD for treated vs. untreated items across all families at d025 MMLU. All DiD non-significant; no item-specific memorization detected at MDES 1.2–2.0pp.

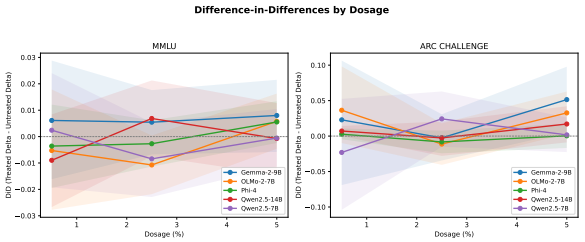


Figure 9: Difference-in-differences (treated vs. untreated items) across contamination dosage levels. DiD remains non-significant at all dosages for all model families, confirming that the global capability modulation mechanism (§5.4) is consistent across the full dosage range.

N DiD Visualization

O DiD Across Dosage Levels

P Post-hoc Power Analysis (MDES)

Table 12 and Table 13 report the minimum detectable effect size (MDES) for the treated vs. untreated DiD comparison at d025 (80% power, $\alpha = 0.05$, two-sided). MDES is computed as the effect size detectable given each model's observed variance, the treated/untreated split sizes, and a two-sample t-test power calculation. At d100, all benchmark items are treated ($n_{\text{untreated}} = 0$), precluding DiD analysis; d025 is reported as the primary dosage level.

Table 12: Post-hoc MDES for MMLU DiD ($n_{\text{treated}} = 3{,}540$, $n_{\text{untreated}} = 10{,}502$).

Model	DiD (pp)	MDES (pp)
Phi-4 14.7B	−0.27	1.25
OLMo-2 7B	−1.07	1.59
Gemma-2 9B	+0.55	1.74
Qwen2.5-7B	−0.84	2.02
Qwen2.5-14B	+0.69	2.05

On MMLU, all observed DiD values are well below their respective MDES, confirming that the null results are consistent with absence of item-specific effects >2pp. On ARC, the smaller treated sample ($n = 257$) yields substantially

Table 13: Post-hoc MDES for ARC-Challenge DiD ($n_{\text{treated}} = 257$, $n_{\text{untreated}} = 915$).

Model	DiD (pp)	MDES (pp)
Phi-4 14.7B	−0.83	2.28
Qwen2.5-14B	−0.26	3.47
OLMo-2 7B	−1.13	4.19
Gemma-2 9B	−0.25	4.66
Qwen2.5-7B	+2.43	5.57

higher MDES; in particular, Qwen-7B's MDES of 5.57pp means that even moderate item-specific effects cannot be excluded for this model on ARC. ARC DiD results should therefore be interpreted as inconclusive rather than as evidence against memorization.

Q Volume Control Analysis

To disentangle data quantity from contamination content, we train Falcon3-7B on 66,533 non-benchmark MCQ items drawn from MMLU auxiliary training subjects, filtered to exclude test-set overlap (MD5 deduplication). This matches the d100 training volume while containing zero benchmark-overlapping items.

Table 14: Volume control decomposition (Falcon3-7B, seed=42). Data quantity contributes 45% (MMLU) and 43% (ARC) of the total Benefit; contamination content is the majority contributor.

Condition	MMLU	ARC	vs. d000
d000 (clean baseline)	58.77%	81.57%	—
Volume control	60.67%	83.28%	+1.90 / +1.71pp
d100 (contaminated)	62.96%	85.58%	+4.19 / +4.01pp

The cross-benchmark consistency of the decomposition (MMLU 45% vs. ARC 43%) strengthens confidence in the estimate. Applying Falcon3's volume baseline to other patterns: Qwen-14B's Collapse (−9.4pp) against a +1.90pp volume baseline implies contamination-specific harm of approximately −11.3pp; Phi-4's Immune response (<1pp) in the presence of volume-driven gains further confirms that contamination vulnerability is family-specific rather than a generic data quantity effect.

R Dose-Response Details

The dose-response patterns reported in §5.1 are measured at the 7B–14.7B scale. Within the Qwen2.5 family at ≤3B, effects are considerably smaller: at 1.5B and 3B, MMLU accuracy increases with contamination dosage by ∼0.6–0.7pp, with diminishing returns above d025. At 0.5B, however, aggregate MMLU accuracy *decreases* with dosage (0.41 at d000 → 0.27 at d100), consistent with the position bias confound documented in §4.1: SFT injection at this scale disrupts the model's general capability, and the resulting A-option bias (up to 98%) overwhelms any memorization benefit.

S Sensitivity of Profile Classification

We assess the robustness of Algorithm 1 to threshold selection via a grid sweep over 134 (τ, σ) combinations. Table 15 reports family-level stability across both a reasonable parameter range ($\tau \in [0.5, 2.0]$ pp × $\sigma \in [3.0, 7.0]$ pp; 30 combinations) and the full 134-combination sweep, along with the nearest parameter value at which the classification transitions.

Table 15: Profile classification stability across threshold parameter ranges. All families achieve 100% stability within the reasonable range; transitions occur only at extreme parameter values (≥3× default).

Family	Pattern	Reasonable	Full	Nearest transition
Qwen2.5-14B	Collapse	100%	100%	$\tau = 9.4$
Gemma-2 9B	Benefit	100%	89%	$\tau = 3.8$
Phi-4 14.7B	Immune	100%	82%	$\tau < 0.35$
OLMo-2 7B	Benefit	100%	82%	$\tau = 3.25$
Qwen2.5-7B	V-recovery	100%	72%	$\tau = 3.0$

Within the reasonable parameter range, all five families achieve 100% classification stability (150/150 family–parameter combinations). Transitions occur only at extreme τ values well beyond the default ($\tau = 2$): Qwen-14B requires $\tau = 9.4$ (4.7× default) to reclassify, reflecting the large separation between its Δacc (−9.4pp) and any plausible noise threshold. Qwen-7B is the least robust at $\tau = 3.0$ (1.5× default), but this is still above any defensible noise threshold for MMLU. The high intrinsic effect-size separation—Phi-4 fluctuates <0.4pp while Qwen-14B declines by 9.4pp—means that classification stability is driven by the data rather than by parameter choice. At $\tau = 1$ pp (the lower bound of the reasonable range), all five original families and the held-out LLaMA-3.1 ($|\Delta| = 0.69$ pp) retain their classifications.

T Cross-Family Domain Decomposition

We decompose MMLU accuracy changes into four constituent domains (STEM, Humanities, Social Sciences, Other) across all model families at

each contamination dosage.

Two patterns emerge. First, the direction of accuracy change is governed by the dose-response pattern, not by domain. At d100, Collapse (Qwen-14B) produces negative Δacc in all four domains (STEM -6.8pp, Humanities -10.1pp, Social Sciences -11.2pp, Other -9.4pp), while Benefit (Gemma-2) produces positive Δacc throughout ($+2.7$, $+3.9$, $+5.0$, $+3.4$pp). Immune (Phi-4) remains within ± 1.0pp across all domains. No pattern produces a crossover exceeding the noise band ($|\Delta| > 1$pp); the single exception is V-recovery Humanities ($+0.28$pp, well within sampling noise). This consistency holds at intermediate dosages as well.

Second, domains differ in sensitivity. STEM is the most resilient (mean $|\Delta\text{acc}| = 2.8$pp across all families at d100). Social Sciences is the most sensitive at 5.1pp—1.8$\times$ the STEM magnitude. The ordering is stable across patterns: in Collapse, STEM shows the smallest decline; in Benefit, STEM shows the smallest gain.

These results reinforce the global capability modulation interpretation (§5.4): contamination direction is determined by the model's dose-response pattern rather than domain-specific memorization. The differential domain sensitivity more likely reflects variation in item difficulty distributions than differential memorization rates.

U Baseline Ability vs. Contamination Effect

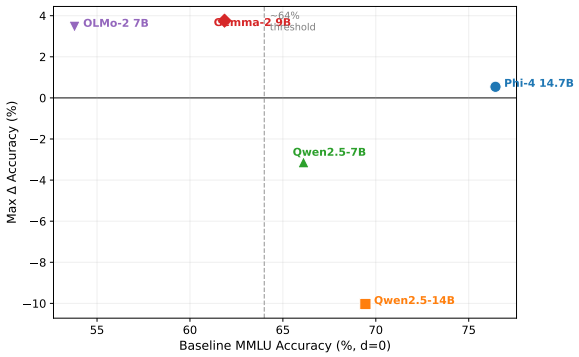


Figure 10: Baseline MMLU accuracy (d000) vs. maximum accuracy change (Δacc) under contamination. A threshold between 61.9% (Gemma-2) and 66.1% (Qwen-7B) separates models that benefit from contamination (below) from those that are harmed or immune (above). Estimated from models spanning four independent families.

V Per-Domain γ Estimates

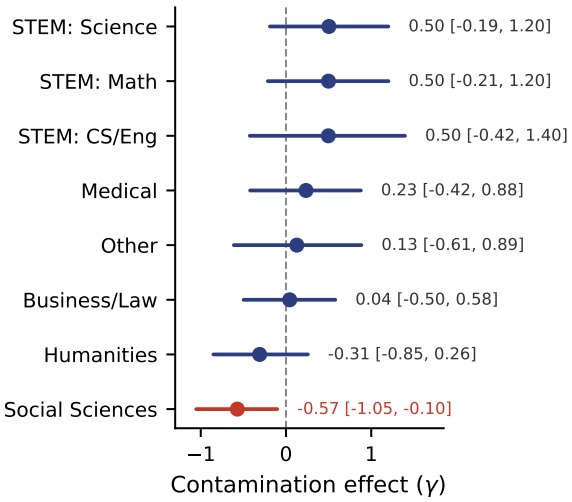


Figure 11: Per-domain γ_d estimates with 95% CIs (J=13 Qwen2.5). STEM domains show positive γ_d; Social Sciences is negative and survives Bonferroni correction at J=16.

Table 16: Per-domain γ estimates (2PL+γ, Qwen2.5) at J=13 and J=16. All J=16 fits converge ($\hat{R} < 1.01$). Social Sciences $P(\gamma < 0)$ strengthens from 99.1% to 99.7% at J=16, exceeding the Bonferroni-corrected threshold (99.375% for eight domains) under baseline and loose priors but falling marginally short under the tight prior (Table 9). Results remain exploratory at these ensemble sizes (Gelman et al., 2012). ARC-Challenge is excluded (PPP < 0.05 at all J).

	J=13		J=16			
Domain	γ	$P(\gamma<0)$	γ	$P(\gamma<0)$	PPP	$\hat{R}$
MMLU: Math	+0.50	8.1%	+0.60	3.8%	0.305	1.007
MMLU: Science	+0.50	7.8%	+0.57	4.3%	0.197	1.007
MMLU: CS/Eng	+0.50	14.8%	+0.32	24.0%	0.503	1.006
MMLU: Medical	+0.24	24.0%	+0.05	43.6%	0.268	1.008
MMLU: Other	+0.13	35.7%	+0.32	20.4%	0.072†	1.008
MMLU: Business/Law	+0.04	44.1%	+0.08	37.8%	0.395	1.007
MMLU: Humanities	−0.31	85.4%	−0.14	69.7%	0.416	1.007
MMLU: Social Sci	−0.57	**99.1%**	−0.63	**99.7%**	0.722	1.006

†PPP below acceptable threshold (0.1); γ estimate for this domain should be interpreted with caution.

W DGP Robustness Analysis

We evaluate 2PL+γ robustness under data-generating process (DGP) misspecification via 400 simulation runs (2 DGPs $\times$ 4 ensemble sizes $\times$ 5 true γ values $\times$ 10 seeds).

Design. The *Linear DGP* (control) generates responses from the correctly specified additive model (Eq. 2). The *Threshold DGP* (misspecified) applies a step-function contamination effect: items with exposure above a threshold receive the

full effect; items below receive zero. Ensemble sizes sweep $J \in \{16, 50, 100, 500\}$; true $\gamma \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$.

Point estimates. Under the Linear DGP, $\hat{\gamma}/\gamma_{\text{true}} = 0.84$–$1.07$ across all conditions (unbiased within sampling noise). Under the Threshold DGP, $\hat{\gamma}/\gamma_{\text{true}} = 0.94$–$1.24$ (mild overestimation, not attenuation). PPP remains in the acceptable range (0.50–0.56) under both DGPs, confirming that posterior predictive checks cannot detect this form of misspecification.

CI coverage. Under the Linear DGP, 95% CI coverage is nominal ($\geq$0.93) at all J. Under the Threshold DGP, coverage degrades with ensemble size (Table 17). The intervals become overconfident as J increases because the model converges to a biased point estimate.

Table 17: 95% CI coverage under Threshold DGP (range across γ values).

J	95% CI coverage
50	1.00
100	0.90–1.00
200	0.80–1.00
500	0.70–1.00

Implications. The reasoning chain is: (i) Threshold DGP + linear model yields $\hat{\gamma}/\gamma_{\text{true}} \geq 0.94$ (overestimation, not attenuation); (ii) therefore, if a true threshold effect exists, the linear model will not fail to detect it; (iii) an empirical $\gamma \approx 0$ is a genuine null, not a misspecification artifact. Coverage degradation affects the *precision* of effect size estimates (how narrow the CI is), not the *detection* of whether an effect exists. At the ensemble sizes in our empirical analyses (J=13–16), coverage remains near-nominal even under misspecification. Real RRF exposure is right-skewed (mean=0.041); the simulation uses uniform exposure, so quantitative magnitudes may differ, but the qualitative pattern (overestimation direction, coverage degradation at large J) is robust.

X ARC IRT Results

ARC-Challenge IRT results are reported separately due to marginal model fit (PPP=0.003 at $J = 16$). The smaller item pool (1,172 vs. 14,042 for MMLU) and more homogeneous difficulty distribution contribute to structural misfit: the 2PL+γ model cannot adequately capture the item response patterns. Under this caveat, the ARC baseline comparison ($J = 16$, 300 RRF items) yields: B0-Raw MAE=0.007, B3-Vanilla2PL MAE=0.015, B4-LinearRegression MAE=0.012 ($\tau = 0.731$, best rank correlation), B6-DCR MAE=0.314 (systematic overcorrection), ContamCal MAE=0.020. The pattern mirrors MMLU: raw accuracy dominates on MAE because the aggregate contamination effect ($\sim$0.7pp) is smaller than correction noise. Domain-level γ estimation on ARC is unreliable at all tested configurations ($J = 8$, 13, 16; PPP < 0.05 in all cases).

Y Full Baseline Comparison

Table 3 (main text) reports B0, B3, and ContamCal under the Qwen2.5 pooled configuration ($J = 16$ for MMLU/ARC, $J = 24$ for GSM8K which retains 0.5B). Table 18 presents the complete seven-baseline comparison on ARC under a pooled 0.5B+1.5B configuration ($J = 8$, S1-only exposure). This configuration differs from the main analysis ($J = 16$, RRF exposure): the smaller ensemble and single-signal exposure limit comparability with MMLU results; these results are included for completeness. Table 19 presents GSM8K baselines at 0.5B scale.

B6-DCR (Xu et al., 2025) produces the highest MAE on both benchmarks, consistent with its fuzzy inference approach, which overcorrects scores without a measurement-theoretic anchor. B4-LinearRegression performs competitively on ARC (MAE=0.056) but lacks the uncertainty quantification and domain-level diagnosis that IRT provides. B1-Remove&Rescore and B5-Emperor do not improve over raw scores, which confirms that item removal and mitigation strategies are insufficient for calibration.

Table 18: Full baseline comparison on ARC-Challenge (pooled 0.5B+1.5B, $J = 8$, S1-only exposure). MAE↓.

Method	ARC MAE
B0-Raw	0.059
B1-Remove&Rescore	0.059
B2-Single2PL+γ	0.064
B3-Vanilla4PL	0.071
B4-LinearRegression	0.056
B5-Emperor	0.059
B6-DCR	—
ContamCal (4PL+γ)	**0.035**

Table 19: Baseline comparison on GSM8K (0.5B scale, delta exposure). MAE↓.

Method	GSM8K MAE
B0-Raw	0.003
B2-Single2PL+γ	0.320
B3-Vanilla4PL	0.012
B4-LinearRegression	**0.003**
B6-DCR	—
ContamCal (4PL+γ)	0.020

Z Parametric Dose-Response Model Comparison

We fit four parametric dose-response models—Constant (intercept-only), Linear, Quadratic, and four-parameter Hill (Hill, 1910; Ritz et al., 2015)—to pooled multi-seed MMLU accuracy trajectories for each family. Table 20 reports AICc and Akaike weights.

Table 20: AICc model selection for dose-response trajectories (pooled multi-seed MMLU). Bold indicates best-fitting model per family; w is the Akaike weight of the best model.

Family	Constant	Linear	Quadratic	Hill	Best (w)
Phi-4 14.7B	−71.42	−72.08	−70.06	**−73.14**	Hill (0.58)
Qwen2.5-14B	49.54	23.40	21.22	**18.28**	Hill (0.55)
Qwen2.5-7B	−12.38	−20.53	−14.52	**−24.97**	Hill (0.77)
Gemma-2 9B	**17.35**	18.89	20.01	18.96	Constant (0.39)
OLMo-2 7B	**25.82**	26.66	27.54	26.47	Constant (0.62)

Parametric model selection with pooled multi-seed data recovers only the most extreme pattern (Collapse, Qwen-14B; Hill $E_{\max} = -43$pp). Two families classified as Benefit by Algorithm 1 (Gemma-2, OLMo-2) are indistinguishable from Constant under AICc—consistent with their cross-seed CIs crossing zero (Table 11). Qwen-7B's V-recovery is structurally inaccessible to monotonic Hill/sigmoid models; Quadratic captures the V-shape but is penalized by AICc (ΔAICc=6.4). Phi-4's Immune classification is reclassified as weak Benefit (Hill $E_{\max} = +2.43$pp, CI $[-2.01, +6.88]$ crossing zero). These discrepancies reflect insufficient statistical power at $n = 3$ seeds rather than contradictions: Algorithm 1 resolves per-seed patterns below the detection threshold of pooled parametric methods, motivating its use for profile construction.

AA Exposure Score Item-Level Classification

Under controlled injection where ground-truth contamination labels are available, we evaluate item-level classification performance of each detection signal and the RRF fusion (§3.2). S1 (embedding similarity) achieves near-perfect discrimination (ROC-AUC 0.93–0.99 across benchmarks and dosage levels), confirming that cosine similarity to the training corpus reliably identifies injected items. RRF fusion achieves ROC-AUC 0.73–0.93, reduced relative to S1 because S2 (Min-K%++, AUC $\approx$ 0.48–0.50) and S4 (Self-Critique, AUC $\approx$ 0.47–0.51) contribute near-random item-level signal. However, RRF is designed for the realistic setting where ground truth is unavailable: S1 requires an explicit training corpus reference, which practitioners typically lack, while RRF's multi-signal approach provides robustness across detection scenarios where no single signal dominates (§E).